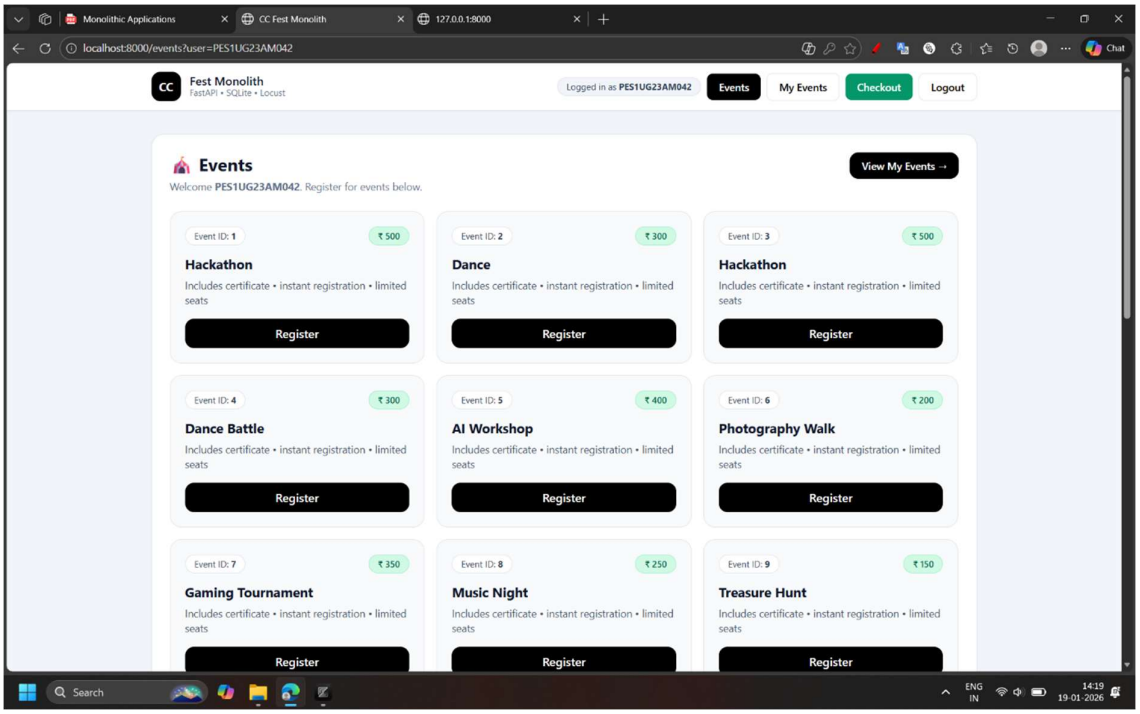
CLOUD COMPUTING LAB-2

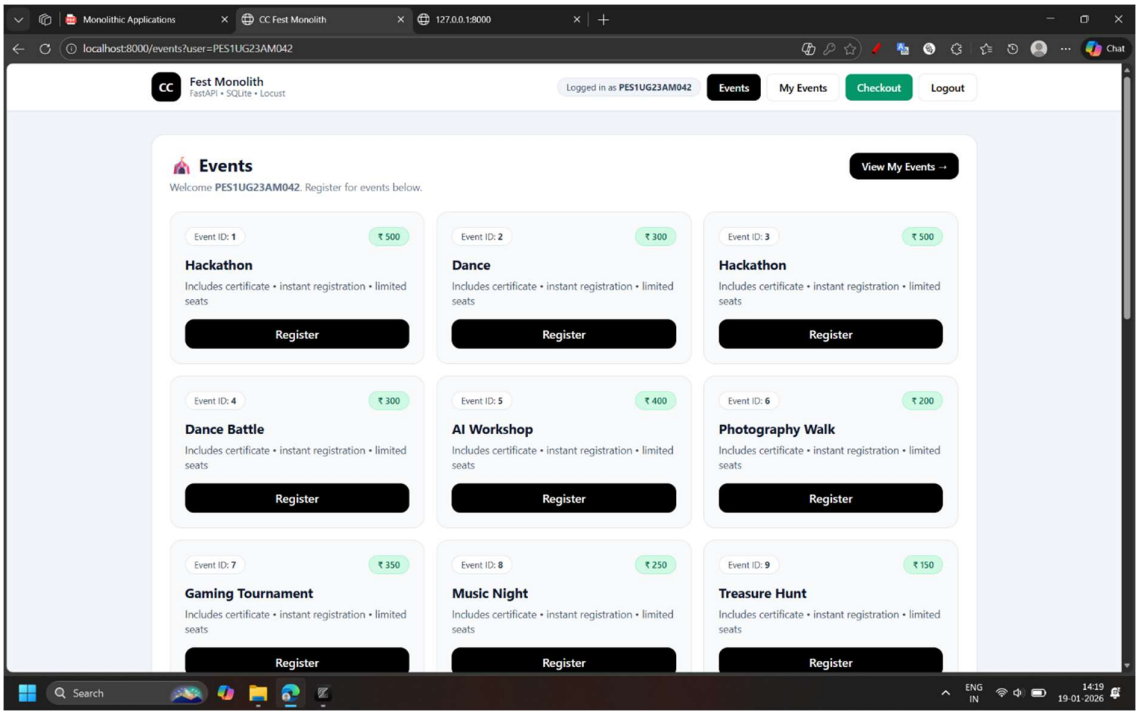
NAME : Amit Prakash

SRN: PES1UG23AM042

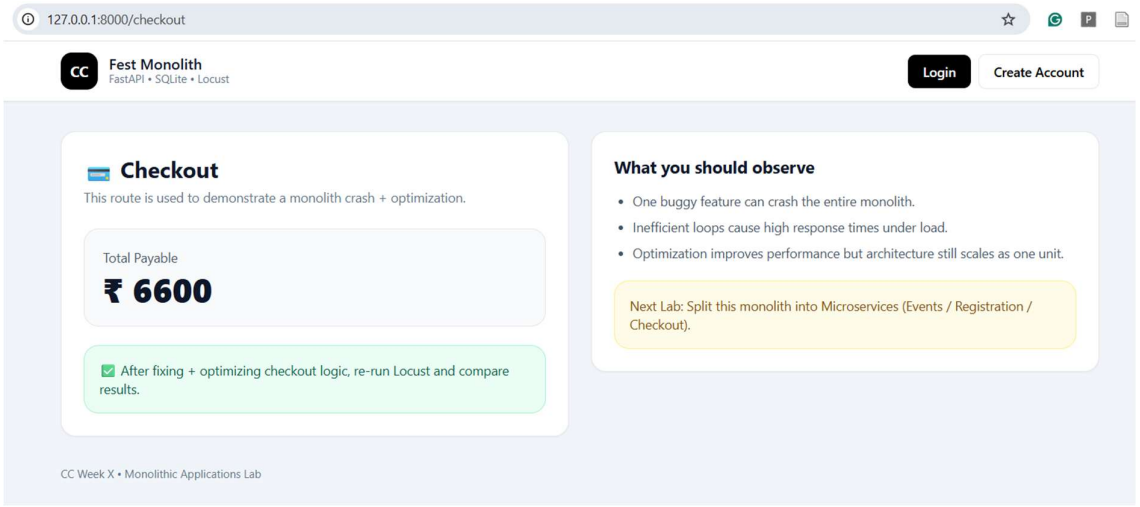
SS-1



SS-2

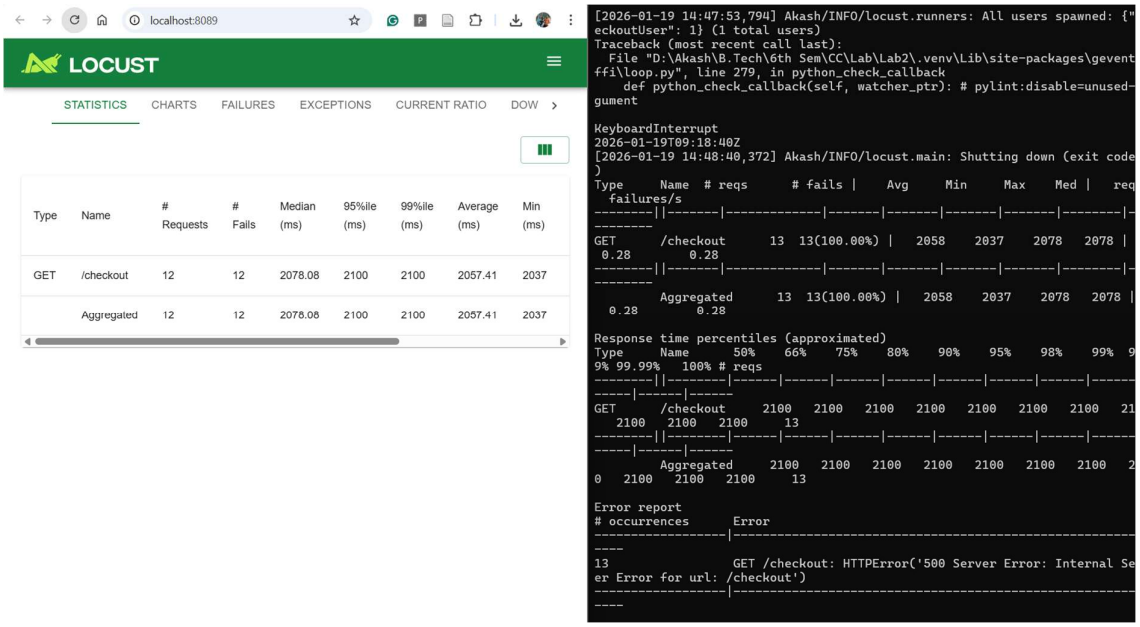

Monolithic Applications
CC Fest Monolith
127.0.0.1:8000
localhost:8000/events?user=PES1UG23AM042
Fest Monolith
FastAPI • SQLite • Locust
Logged in as PES1UG23AM042
Events
My Events
Checkout
Logout
Events
View My Events →
Welcome PES1UG23AM042. Register for events below.
Event ID: 1
₹ 500
Hackathon
Includes certificate • instant registration • limited seats
Register
Event ID: 2
₹ 300
Dance
Includes certificate • instant registration • limited seats
Register
Event ID: 3
₹ 500
Hackathon
Includes certificate • instant registration • limited seats
Register
Event ID: 4
₹ 300
Dance Battle
Includes certificate • instant registration • limited seats
Register
Event ID: 5
₹ 400
AI Workshop
Includes certificate • instant registration • limited seats
Register
Event ID: 6
₹ 200
Photography Walk
Includes certificate • instant registration • limited seats
Register
Event ID: 7
₹ 350
Gaming Tournament
Includes certificate • instant registration • limited seats
Register
Event ID: 8
₹ 250
Music Night
Includes certificate • instant registration • limited seats
Register
Event ID: 9
₹ 150
Treasure Hunt
Includes certificate • instant registration • limited seats
Register
Search
ENG
IN
14:19
19-01-2026

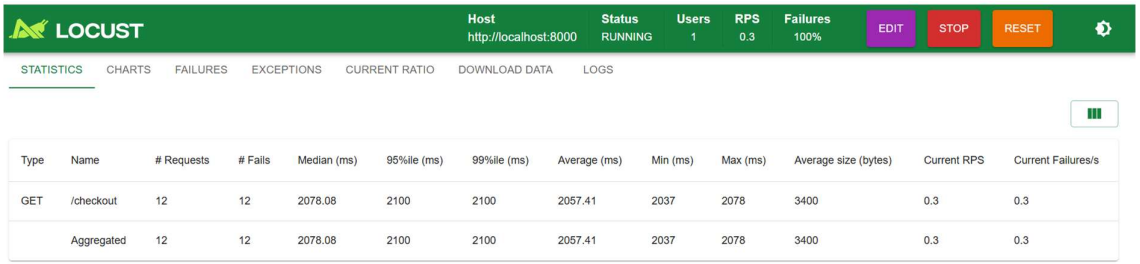
SS3


127.0.0.1:8000/checkout
Fest Monolith
FastAPI • SQLite • Locust
Login
Create Account
Checkout
This route is used to demonstrate a monolith crash + optimization.
Total Payable
₹ 6600
After fixing + optimizing checkout logic, re-run Locust and compare results.
What you should observe
One buggy feature can crash the entire monolith.
Inefficient loops cause high response times under load.
Optimization improves performance but architecture still scales as one unit.
Next Lab: Split this monolith into Microservices (Events / Registration / Checkout).
CC Week X • Monolithic Applications Lab

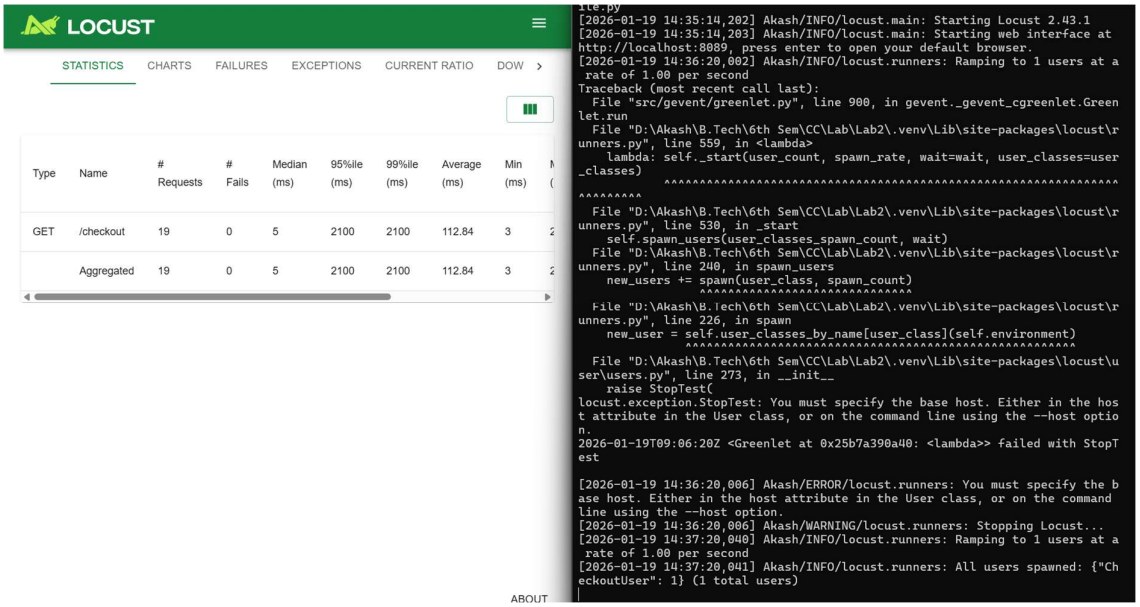
INFO: 127.0.0.1:64945 - "GET /checkout HTTP/1.1" 200 OK

SS4:


localhost:8089
LOCUST
STATISTICS
CHARTS
FAILURES
EXCEPTIONS
CURRENT RATIO
DOW
Type | Name | # Requests | # Fails | Median (ms) | 95%ile (ms) | 99%ile (ms) | Average (ms) | Min (ms)
GET | /checkout | 12 | 12 | 2078.08 | 2100 | 2100 | 2057.41 | 2037
Aggregated | 12 | 12 | 2078.08 | 2100 | 2100 | 2057.41 | 2037
[2026-01-19 14:47:53,794] Akash/INFO/locust.runners: All users spawned: {"CheckoutUser": 1} (1 total users)
Traceback (most recent call last):
File "D:\Akash\B.Tech\6th Sem\CC\Lab\Lab2\.venv\Lib\site-packages\gevent\ffi\loop.py", line 279, in python_check_callback
def python_check_callback(self, watcher_ptr): # pylint:disable=unused-argument
KeyboardInterrupt
2026-01-19T09:18:40Z
[2026-01-19 14:48:40,372] Akash/INFO/locust.main: Shutting down (exit code)
Type Name # reqs # fails | Avg Min Max Med | req/s failures/s
GET /checkout 13 13(100.00%) | 2058 2037 2078 2078 | 0.28 0.28
Aggregated 13 13(100.00%) | 2058 2037 2078 2078 | 0.28 0.28
Response time percentiles (approximated)
Type Name 50% 66% 75% 80% 90% 95% 98% 99% 99.9% 99.99% 100% # reqs
GET /checkout 2100 2100 2100 2100 2100 2100 2100 2100 2100 2100 2100 13
Aggregated 2100 2100 2100 2100 2100 2100 2100 2100 2100 2100 2100 13
Error report
occurrences Error
13 GET /checkout: HTTPError('500 Server Error: Internal Server Error for url: /checkout')


LOCUST
Host
http://localhost:8000
Status
RUNNING
Users
1
RPS
0.3
Failures
100%
EDIT
STOP
RESET
STATISTICS
CHARTS
FAILURES
EXCEPTIONS
CURRENT RATIO
DOWNLOAD DATA
LOGS
Type | Name | # Requests | # Fails | Median (ms) | 95%ile (ms) | 99%ile (ms) | Average (ms) | Min (ms) | Max (ms) | Average size (bytes) | Current RPS | Current Failures/s
GET | /checkout | 12 | 12 | 2078.08 | 2100 | 2100 | 2057.41 | 2037 | 2078 | 3400 | 0.3 | 0.3
Aggregated | 12 | 12 | 2078.08 | 2100 | 2100 | 2057.41 | 2037 | 2078 | 3400 | 0.3 | 0.3

SS5


LOCUST
STATISTICS
CHARTS
FAILURES
EXCEPTIONS
CURRENT RATIO
DOW
Type | Name | # Requests | # Fails | Median (ms) | 95%ile (ms) | 99%ile (ms) | Average (ms) | Min (ms)
GET | /checkout | 19 | 0 | 5 | 2100 | 2100 | 112.84 | 3
Aggregated | 19 | 0 | 5 | 2100 | 2100 | 112.84 | 3
ABOUT
[2026-01-19 14:35:14,202] Akash/INFO/locust.main: Starting Locust 2.43.1
[2026-01-19 14:35:14,203] Akash/INFO/locust.main: Starting web interface at http://localhost:8089, press enter to open your default browser.
[2026-01-19 14:36:20,002] Akash/INFO/locust.runners: Ramping to 1 users at a rate of 1.00 per second
Traceback (most recent call last):
File "src/gevent/greenlet.py", line 900, in gevent._gevent_cgreenlet.Greenlet.run
File "D:\Akash\B.Tech\6th Sem\CC\Lab\Lab2\.venv\Lib\site-packages\locust\runners.py", line 559, in <lambda>
lambda: self._start(user_count, spawn_rate, wait=wait, user_classes=user_classes)
File "D:\Akash\B.Tech\6th Sem\CC\Lab\Lab2\.venv\Lib\site-packages\locust\runners.py", line 530, in _start
self.spawn_users(user_classes_spawn_count, wait)
File "D:\Akash\B.Tech\6th Sem\CC\Lab\Lab2\.venv\Lib\site-packages\locust\runners.py", line 240, in spawn_users
new_users += spawn(user_class, spawn_count)
File "D:\Akash\B.Tech\6th Sem\CC\Lab\Lab2\.venv\Lib\site-packages\locust\runners.py", line 226, in spawn
new_user = self.user_classes_by_name[user_class](self.environment)
File "D:\Akash\B.Tech\6th Sem\CC\Lab\Lab2\.venv\Lib\site-packages\locust\user\users.py", line 273, in __init__
raise StopTest(
locust.exception.StopTest: You must specify the base host. Either in the host attribute in the User class, or on the command line using the --host option.
2026-01-19T09:06:20Z <Greenlet at 0x25b7a390a40: <lambda>> failed with StopTest
[2026-01-19 14:36:20,006] Akash/ERROR/locust.runners: You must specify the base host. Either in the host attribute in the User class, or on the command line using the --host option.
[2026-01-19 14:36:20,006] Akash/WARNING/locust.runners: Stopping Locust...
[2026-01-19 14:37:20,040] Akash/INFO/locust.runners: Ramping to 1 users at a rate of 1.00 per second
[2026-01-19 14:37:20,041] Akash/INFO/locust.runners: All users spawned: {"CheckoutUser": 1} (1 total users)

```
Type     Name                                          # reqs      # fails |    Avg
    Min     Max    Med |   req/s  failures/s
--------|-----------------------------------|-------|-------------|-------|
-------|-------|-------|--------|-----------
GET      /checkout                                        19     0(0.00%) |    112
      3    2050      5 |    0.65        0.00
--------|-----------------------------------|-------|-------------|-------|
-------|-------|-------|--------|-----------
         Aggregated                                       19     0(0.00%) |    112
      3    2050      5 |    0.65        0.00

Response time percentiles (approximated)
Type     Name                                                 50%    66%    75%
80%    90%    95%    98%    99%  99.9% 99.99%   100% # reqs
--------|-----------------------------------------|--------|------|------|---
---|------|------|------|------|------|------|------|------
GET      /checkout                                          5      5      6
  7     12   2100   2100   2100   2100   2100   2100     19
--------|-----------------------------------------|--------|------|------|---
---|------|------|------|------|------|------|------|------
         Aggregated                                         5      5      6
  7     12   2100   2100   2100   2100   2100   2100     19
```

LOCUST

Host	Status	RPS	Failures
http://localhost:8000	STOPPED	0.6	0%

NEW RESET

STATISTICS CHARTS FAILURES EXCEPTIONS CURRENT RATIO DOWNLOAD DATA LOGS

Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/checkout	17	0	10	2100	2100	135.35	7	2108	2797	0.6	0
	Aggregated	17	0	10	2100	2100	135.35	7	2108	2797	0.6	0

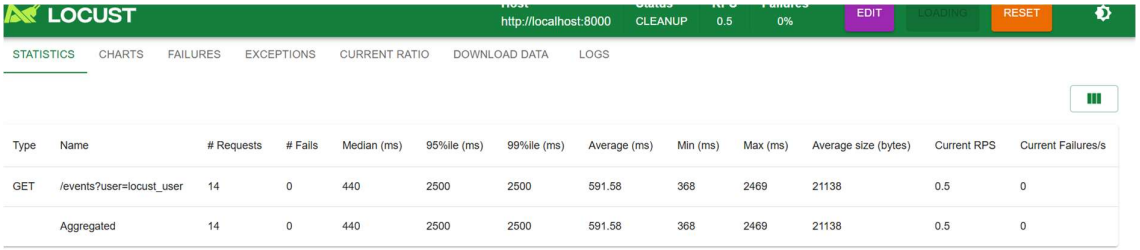
Requests per second before Optimization: 0.3

RPS after Optimization: 0.6

Average Time After Optimization: 135.35 ms

Average Time Before Optimization: 2057.41 ms

SS - 6

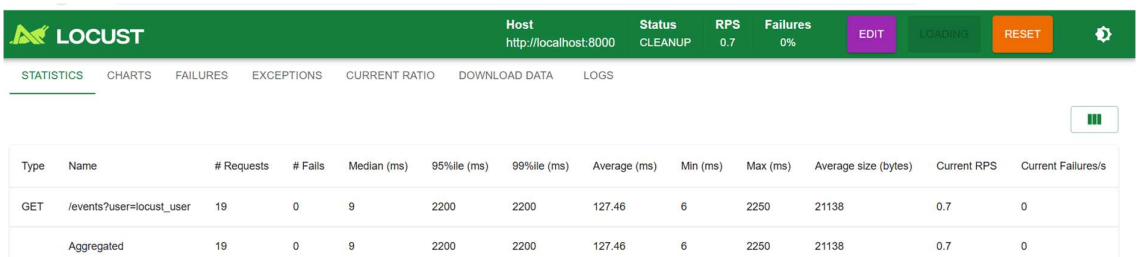


Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/events?user=locust_user	14	0	440	2500	2500	591.58	368	2469	21138	0.5	0
	Aggregated	14	0	440	2500	2500	591.58	368	2469	21138	0.5	0

```
Type     Name                                                                          # reqs      # fails |    Avg     Min     Max    Med |   req/s  failures/s
--------|------------------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
GET      /events?user=locust_user                                                          15     0(0.00%) |    593     368    2469    450 |    0.50        0.00
--------|------------------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
         Aggregated                                                                        15     0(0.00%) |    593     368    2469    450 |    0.50        0.00

Response time percentiles (approximated)
Type     Name                                                                             50%    66%    75%    80%    90%    95%    98%    99%  99.9% 99.99
%   100% # reqs
--------|------------------------------------------------------------------------------|--------|------|------|------|------|------|------|------|------|-----
-|------|------
GET      /events?user=locust_user                                                         450    470    520    530    630   2500   2500   2500   2500   250
0   2500     15
--------|------------------------------------------------------------------------------|--------|------|------|------|------|------|------|------|------|-----
-|------|------
         Aggregated                                                                       450    470    520    530    630   2500   2500   2500   2500   250
0   2500     15
```

SS -7

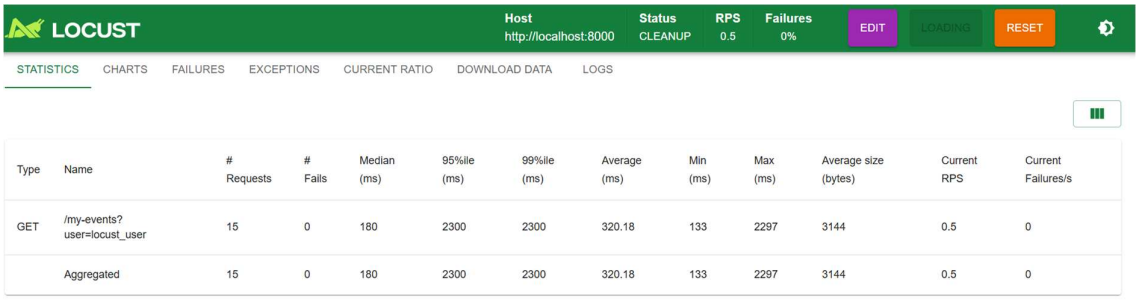


Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/events?user=locust_user	19	0	9	2200	2200	127.46	6	2250	21138	0.7	0
	Aggregated	19	0	9	2200	2200	127.46	6	2250	21138	0.7	0

```
2026-01-19T09:45:45Z
[2026-01-19 15:15:45,519] Akash/INFO/locust.main: Shutting down (exit code 0)
Type     Name                                                                          # reqs      # fails |    Avg     Min     Max    Med |   req/s  failures/s
--------|------------------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
GET      /events?user=locust_user                                                          19     0(0.00%) |    127       6    2249      9 |    0.65        0.00
--------|------------------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
         Aggregated                                                                        19     0(0.00%) |    127       6    2249      9 |    0.65        0.00

Response time percentiles (approximated)
Type     Name                                                                             50%    66%    75%    80%    90%    95%    98%    99%  99.9% 99.99
%   100% # reqs
--------|------------------------------------------------------------------------------|--------|------|------|------|------|------|------|------|------|-----
-|------|------
GET      /events?user=locust_user                                                           9     10     11     12     15   2200   2200   2200   2200   220
0   2200     19
--------|------------------------------------------------------------------------------|--------|------|------|------|------|------|------|------|------|-----
-|------|------
         Aggregated                                                                         9     10     11     12     15   2200   2200   2200   2200   220
0   2200     19
```

SS -8

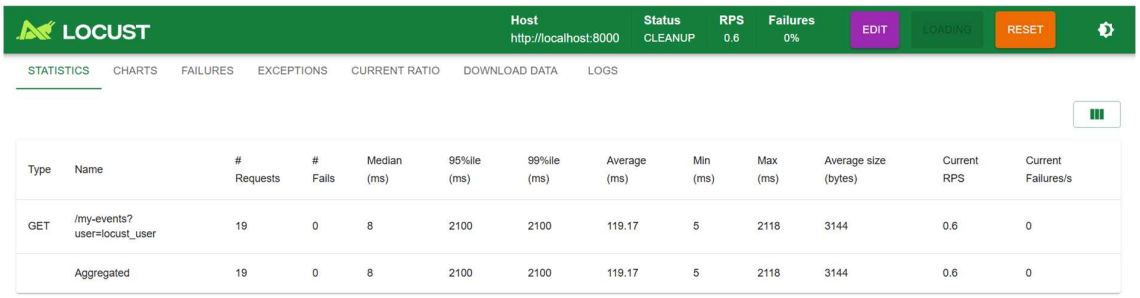


Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/my-events?user=locust_user	15	0	180	2300	2300	320.18	133	2297	3144	0.5	0
	Aggregated	15	0	180	2300	2300	320.18	133	2297	3144	0.5	0

```
[2026-01-19 15:19:43,870] Akash/INFO/locust.main: Shutting down (exit code 0)
Type     Name                                                          # reqs      # fails |    Avg     Min     Max    Med |   req/s  failures/s
--------|-----------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
GET      /my-events?user=locust_user                                       15     0(0.00%) |    320     133    2296    180 |    0.53        0.00
--------|-----------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
         Aggregated                                                        15     0(0.00%) |    320     133    2296    180 |    0.53        0.00

Response time percentiles (approximated)
Type     Name                                                                  50%    66%    75%    80%    90%    95%    98%    99%  99.9% 99.99
%   100% # reqs
--------|-----------------------------------------------------------------------|--------|------|------|------|------|------|------|------|------|-----
-|------|------
GET      /my-events?user=locust_user                                           180    190    200    230    230   2300   2300   2300   2300    230
0   2300     15
--------|-----------------------------------------------------------------------|--------|------|------|------|------|------|------|------|------|-----
-|------|------
         Aggregated                                                            180    190    200    230    230   2300   2300   2300   2300    230
0   2300     15
```

SS – 9



Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/my-events?user=locust_user	19	0	8	2100	2100	119.17	5	2118	3144	0.6	0
	Aggregated	19	0	8	2100	2100	119.17	5	2118	3144	0.6	0

```
[2026-01-19 15:21:31,236] Akash/INFO/locust.main: Shutting down (exit code 0)
Type     Name                                                          # reqs      # fails |    Avg     Min     Max    Med |   req/s  failures/s
--------|-----------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
GET      /my-events?user=locust_user                                       19     0(0.00%) |    119       4    2118      8 |    0.66        0.00
--------|-----------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
         Aggregated                                                        19     0(0.00%) |    119       4    2118      8 |    0.66        0.00

Response time percentiles (approximated)
Type     Name                                                                  50%    66%    75%    80%    90%    95%    98%    99%  99.9% 99.99
%   100% # reqs
--------|-----------------------------------------------------------------------|--------|------|------|------|------|------|------|------|------|-----
-|------|------
GET      /my-events?user=locust_user                                             8      9     10     10     12   2100   2100   2100   2100    210
0   2100     19
--------|-----------------------------------------------------------------------|--------|------|------|------|------|------|------|------|------|-----
-|------|------
         Aggregated                                                              8      9     10     10     12   2100   2100   2100   2100    210
0   2100     19
```

For both the routes:

1. **What was the bottleneck?**

   The bottleneck was that there were for loops in the code which were not doing any work and had a very high number like in lakhs which was causing issues

2. **What change did you make?**

   Commented both the for loops to increase the efficiency.

3. **Why did the performance improve?**
   The performance improved because there were some unwanted code blocks which were using resources and doing no work and it was consuming time so if removed perform very efficiently.

Github Repository:

Shared Repository with contents from Lab1 and Lab2 in separated directories

https://github.com/WazupSteve/cloud-computing-lab/tree/main/lab2/PES1UG23AM042